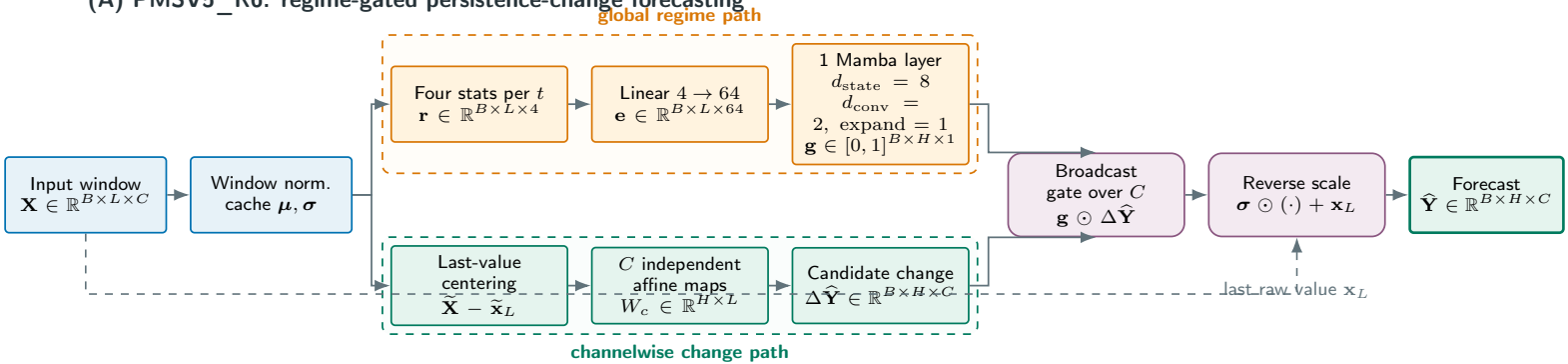
(A) PMSV5_R6: regime-gated persistence-change forecasting



(B) Forward equations

Global regime gate

$$\mathbf{r}_t = [\mathrm{mean}_c(\widetilde{x}_{t,c}), \mathrm{std}_c(\widetilde{x}_{t,c}), \mathrm{mean}_c(\Delta\widetilde{x}_{t,c}), \mathrm{std}_c(\Delta\widetilde{x}_{t,c})]$$

$$\mathbf{h}_{1:L} = \mathrm{Mamba}(W_r \mathbf{r}_{1:L}), \qquad \mathbf{g} = \mathrm{sigmoid}(W_g \mathbf{h}_L + \mathbf{b}_g)$$

Channelwise persistence-change forecast

$$\Delta\widehat{\mathbf{Y}}_c = W_c(\widetilde{\mathbf{X}}_c - \widetilde{x}_{L,c}\mathbf{1}_L) + \mathbf{b}_c$$

$$\widehat{\mathbf{Y}} = \mathbf{x}_L + \boldsymbol{\sigma} \odot (\mathbf{g} \odot \Delta\widehat{\mathbf{Y}})$$

$\mathbf{g}$ *is shared across channels;* W_c *is channel-specific.*